

Characterizing NGC 6383 using Gaia DR3

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ABSTRACT

This study focuses on determining the characteristics of the young open cluster NGC 6383. To achieve this, the HDBSCAN clustering algorithm is utilized to identify potential cluster members based on galactic coordinates, proper motions, and parallaxes from *Gaia Data Release 3*. Various parameters of NGC 6383, such as tidal radius, cluster radius, core radius, distance through parallax and isochrone-fitting, proper motion, age, metallicity, and relevant others, are assessed. To perform this analysis, we utilize an extension of Hamiltonian Monte Carlo, the No-U-Turn Sampler. The results of this analysis point out that NGC 6383 is a very young open cluster ($\sim 2.7 - 4$ Myr). Furthermore, our investigation reveals that the central binary HD 159176 is not a member of the cluster, which was not discussed in previous studies, leading to a revision of its age and other characteristics. Overall, this study contributes to a more accurate and detailed understanding of NGC 6383's properties.

Key words: open clusters and associations: individual — galaxies: star clusters: general — stars: distances — techniques: photometric — parallaxes — proper motions

1 INTRODUCTION

The precise identification of cluster membership is a critical step towards elucidating the physical attributes of a cluster, including its age, distance, and chemical composition. Various methodologies have been employed to determine the membership of sources, such as selection via the Color-Magnitude diagram (CMD), proper motions, radial velocities, parallax, and more recently, Bayesian analysis. The application of these diverse techniques often yields a spectrum of values from different studies, occasionally leading to consensus, but at times exposing discrepancies and inaccuracies in the methodologies employed by previous researchers (Trumpler 1930; Zug 1937; Sanford 1949; Aidelman et al. 2018; Fitzgerald et al. 1978; Pandey et al. 1989; Battinelli & Capuzzo-Dolcetta 1991; Kharchenko et al. 2005; Paunzen et al. 2007; Lindoff 1968).

1.1 NGC 6383

NGC 6383¹ is a young open cluster situated in the Carina-Sagittarius arm, within the *Sh 2-012*² star formation region. It forms part of the larger Sagittarius OB1 association, along with NGC 6530 and NGC 6531 (Rauw & De Becker 2008). The equatorial coordinates (ICRS) of the cluster are $\alpha_{2015.5} = 17^h 34^m 41.5$ and $\delta_{2015.5} = -32^\circ 34' 23''$ (Cantat-Gaudin & Anders 2020), and the galactic coordinates are $\ell = 355.68$ and $b = 0.05$ (Conrad et al. 2017). The

cluster exhibits a radial velocity of -1.2 ± 4.9 km s⁻¹ (Conrad et al. 2017) and a proper motion of $\mu_\alpha = 2.596 \pm 0.018$ mas yr⁻¹ and $\mu_\delta = -1.708 \pm 0.016$ mas yr⁻¹ (Cantat-Gaudin & Anders 2020).

The distance to the cluster has been estimated by various authors, with an upper limit of 2130 pc proposed by Trumpler (1930) and Zug (1937), and a lower limit of 760 pc and 834 pc suggested by Sanford (1949) and Aidelman et al. (2018), respectively. The cluster spans an angular size of 20 arcmin (Morales et al. 2013) and is estimated to be between 1.7 to 5 Myr old (Fitzgerald et al. 1978; Pandey et al. 1989; Battinelli & Capuzzo-Dolcetta 1991; Kharchenko et al. 2005; Paunzen et al. 2007). However, Lindoff (1968) calculated an age of 20 Myr. The reddening $E(B - V) = 0.32 \pm 0.02$ derived by Rauw & De Becker (2008) aligns with previous authors such as Becker & Fenkart (1971); Fitzgerald et al. (1978); Lloyd Evans (1978); The et al. (1985); Pandey et al. (1989); Feinstein (1994); Paunzen et al. (2007), but contradicts Aidelman et al. (2018) who determined a reddening of $E(B - V) = 0.51 \pm 0.03$. This reddening is attributed to a surrounding large shell-structured ionized HII region with a radius of approximately 1 deg, which was likely ionized by its central star, HD 159176.

HD 159176, a contact binary system composed of O7V type stars and a well-studied X-ray double-line binary, is responsible for ionizing the HII region. The estimated age of HD 159176 is around 2.3–2.8 Myr (Rauw et al. 2010). A careful analysis of both the ages of the cluster and its central star system suggests that the central star predates the cluster itself. Furthermore, Fitzgerald et al. (1978) proposed that HD 159176 initiated star formation in the cluster's core and beyond.

In the study by Aidelman et al. (2018), it is hypothesized that if the stars numbered as *Star No. 6* are considered to have started to evolve and if HD 159176 is a blue straggler, then the age of the cluster could be between 6 and 10 Myr.

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¹ NGC 6383, also known as NGC 6374 in the New General Catalog and classified as Collinder 334 and Collinder 335 in the Collinder catalog, was initially misclassified in the original Collinder catalog. (Collinder 1931)

² Also known as RCW 132 (Rodgers et al. 1960) or Gum 67

The parameters of the cluster exhibit a wide range of results, contingent on the methodology utilized by the authors in their research. The cluster hosts sources that have recently embarked on the main sequence track, known as Pre-Main Sequence Stars (PMS). Consequently, the age of the cluster has been a topic of considerable debate.

A summary of the historical results can be seen in 1

2 METHODOLOGY

2.1 Data and Quality process

2.1.1 Data Acquisition

The data set used in this study was obtained from the third data release of the Gaia mission (DR3), which was made publicly available in early 2022 (Gaia Collaboration et al. 2016, 2022). This data set contains comprehensive astrometric solutions, including positions, parallax, and proper motion for approximately 1.46 billion sources, with a bright limit of $G \approx 3$ and a limiting magnitude of $G \approx 21$. The use of this data set provides numerous advantages, including high accuracy and precision in measurements, large sample sizes, and a wide range of astrophysical applications.

To acquire the data, a cone search with a radius of 25 arcmin was performed on the Gaia Archive using the name resolved by Sesame Strasbourg (Simbad-NED-VizieR). The search returned a total of 71847 sources.

Furthermore, for the study of infrared stars, we obtained data from Two Micron All-Sky Survey (2MASS) (Skrutskie et al. 2006). This survey contains magnitude on the J -band ($1.235 \mu\text{m}$), H -band ($1.662 \mu\text{m}$), and K_s -band ($2.159 \mu\text{m}$). A cone search with the same 25 arcmin radius was performed on the NASA/IPAC Infrared Science Service with the 2MASS All-Sky Point Source Catalog (PSC), using the name resolved by NED-Simbad, which returned a total of 66500 sources.

2.1.2 Data Filtering

The dataset was filtered to include only sources with distances within the range of 500 pc to 2500 pc, corresponding to the upper and lower distance limits reported in the literature (Section ??). This was accomplished by selecting sources with parallax measurements between 0.4 mas and 2 mas. Sources with negative parallax measurements were excluded, as these may be the result of uncertainties in the Gaia DR3 data, as suggested Luri et al. (2018).

Only sources with parallax measurement uncertainties below 0.5 mas were considered, resulting in a subset of 14002 sources. A small fraction of these sources may have poor astrometric solutions, and the reported values and uncertainties may not be reliable. Therefore, following the *astrometric fidelity* parameter provided by Rybizki et al. (2022), sources with a value of *astrometric fidelity* greater than 0.5 were selected, leaving a final subset of 10951 sources.

This *astrometric fidelity* parameter, is a metric that takes into account various aspects of the astrometric solution. It provides a measure of the reliability of the astrometric solution for a given source, and it was calculated using a neural network that was trained using 17 pertinent Gaia catalog entries and information, as described in detail in the respective paper.

Among the 10951 sources, it is important to select only the data that has valid parameters, in special the position (α, δ, ℓ, b), proper motions ($\mu_{\alpha*}, \mu_{\delta}$), parallax (ϖ) and the magnitudes (G, G_{BP}, G_{RP}), only 10867 of the sources have valid parameters.

Figures/probabilities.pdf

Figure 1. This figure presents the distribution of probabilities for the sources assigned to NGC 6383. The plot shows the Gaia magnitude (G_{mag}) and the probability assigned to each source. The potential members of NGC 6383 are in blue, while the field stars are in orange. Additionally, noise sources are shown as gray scatter points. The dashed horizontal line at a probability threshold of 0.8 is included for reference. The x-axis is G_{mag} , and the y-axis is the probability.

2.1.3 Crossmatch with 2MASS

A cross-match was performed with the remaining sources, to obtain a final list of sources. Gaia DR3 provides pre-computed crossmatch tables for a variety of catalogues, as explained in Marrese et al. (2022). The table used for the crossmatch was `tmass_psc_xsc_best_neighbour`, giving 6748 crossmatched sources.

The J , H , and K_s magnitudes has been added to only those sources with a separation distance of less than 0.5 arcsec to obtain a final list of sources. This process resulted in a total of 6587 sources with 2MASS magnitudes. The join type was only adding the available 2MASS data to the Gaia data, keeping the original 10867 sources.

2.1.4 Systematic Gaia DR3 Parallax Offset

The Gaia DR3 parallax measurements exhibit a systematic offset as discussed in Lindegren et al. (2021), and the zero-point correction for this offset was calculated using the `Gaiadr3_zeropoint` package, which provides analytical functions to estimate the expected parallax zero-point based on factors such as ecliptic latitude, magnitude, and color for any Gaia (e)DR3 source.

To obtain the corrected parallax, we apply the following formula:

$$\varpi_{\text{corr},i} = \varpi_i - Z(x_i)$$

where $\varpi_{\text{corr},i}$ represents the corrected parallax for the i -th source, ϖ_i denotes the Gaia parallax for the i -th source, and $Z(x_i)$ represents the zero-point correction calculated based on the ecliptic latitude, magnitude, and color of the i -th source.

It should be noted that for sources whose apparent magnitude falls

Reference	Members	Core Radius	Tidal Radius	Parallax	Distance	E(B-V)	$(m_v - M_v)_0$	Age
Trumpler (1930)					2130	...	...	...
Zug (1937)					2130	...	...	...
Sanford (1949)					760	...	...	...
Bok et al. (1953)								
Markarian (1954)								
(Sharpless 1959)								
Rodgers et al. (1960)								
Eggen (1961)					1259	...	...	...
Rubin et al. (1962)								
The (1965)								
Eggen (1965)								
The (1966)								
Graham (1967)					1380	...	10.68 ± 0.54	...
Evans (1968)								
Lindoff (1968)					1250	...	10.5	~ 20
Becker & Fenkart (1971)					1065	0.26	10.92	...
Antalová (1972)								
Tovmassian et al. (1973)								
Vogt & Moffat (1975)								
Fitzgerald et al. (1978)					1500 ± 200	0.33 ± 0.02	10.85	1.6 ± 0.4
Lloyd Evans (1978)					1350	0.35	10.6	...
The et al. (1985)					1400 ± 150	0.30 ± 0.01	...	...
Pandey et al. (1989)					...	0.35	11.65	~ 4
Battinelli & Capuzzo-Dolcetta (1991)					1380	...	...	~ 4
Stickland et al. (1993)								
Feinstein (1994)					1400	0.33	...	...
Rastorguev et al. (1999)					1180	...	...	...
van den Ancker et al. (2000)								
Rauw et al. (2003)								
Zwintz et al. (2004, 2005)								
Kharchenko et al. (2005)					985	...	...	~ 5
Paunzen et al. (2007)					1700	0.28	...	4
Piskunov et al. (2008)					...	0.30	10.897	...
Rauw & De Becker (2008)					1300 ± 100	0.32 ± 0.02	...	...
Rauw et al. (2010)								
Aidelman et al. (2018)					834 ± 158	0.51 ± 0.03	9.61 ± 0.38	$\sim 3 - 10$
Kalari (2019)								

Table 1.

outside the expected range of (6 – 21 mag), the offset correction is applied using the values at either 6 or 21 mag. Similarly, sources with pseudocolor outside the expected range of (1.24 – 1.72 mag) receive the correction corresponding to the values at 1.24 or 1.72 mag. It is important to mention that beyond these ranges, no further interpolation is performed.

For simplicity, we will refer to the corrected parallax $\varpi_{\text{corr},i}$ as the parallax ϖ in the subsequent analysis.

2.1.5 Gaia Proper Motion

In this subsection, we describe the correction applied to the proper motions data to account for the rotational differences between the reference frames of bright and faint sources in the Gaia EDR3 catalog³. The correction is necessary to align the proper motions of bright stars with the International Celestial Reference Frame (ICRF).

The study conducted by Cantat-Gaudin & Brandt (2021) revealed that the reference frame of proper motions for bright Gaia EDR3 sources ($G < 13$) exhibits a rotation relative to that of the faint

EDR3 sources. This rotation introduces a proper motion bias, which can reach up to $80 \mu\text{mas yr}^{-1}$ [REVISAR VALOR EN PAPER] in the magnitude range $G = 11 - 13$. To mitigate this bias and align the proper motions of bright sources with the ICRF, a magnitude-based correction was derived.

The correction involves computing the difference between the observed proper motions and the expected proper motions based on the spin parameters listed in the study.

Applying this correction to the proper motions data ensures that the proper motions of bright stars are aligned with the ICRF and reduces the bias introduced by the rotational differences. The corrected proper motions can then be used for further analysis, such as determining kinematic properties, membership in stellar clusters, or studying the dynamics of the Milky Way.

It is important to note that although the magnitude-based correction effectively aligns the proper motions, there may still be residual systematics of approximately 10 mas yr^{-1} related to color dependence. Therefore, caution should be exercised when interpreting the proper motion data, particularly when considering sources with a wide range of colors.

³ The astrometry and photometry for the sources in the full DR3 is exactly the same.

Figure 2. This scatter plot illustrates the relationship between parallax and distance for NGC 6383. The distribution of parallax values is presented as a histogram (orange) and a kernel density estimate (violet), highlighting the underlying density of measurements. Similarly, the distribution of distance estimates is shown as a histogram (orange) and a kernel density estimate (violet). The mean parallax (μ_{ϖ}) is represented by the dashed blue line, along with its uncertainty. The mean distance (d) is depicted by the dashed blue line, also with its associated uncertainty. The data points of NGC 6383 sources are displayed as scattered points, where their parallax is plotted against their G_{mag} magnitude. The color of each point indicates the probability assigned to the source. The green star marker and error bar represent the parallax and G_{mag} magnitude of HD 159176.

2.2 Membership Determination

To identify potential members of the stellar cluster NGC 6383, we employed the HDBSCAN clustering algorithm in this study.

HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that can identify clusters of varying densities and sizes in a dataset. This algorithm constructs a hierarchical representation of the data, where each level represents a different level of clustering granularity. It starts by computing the mutual reachability distance between points in the dataset, which measures the minimum density required for two points to be considered neighbors. This allows the algorithm to identify regions of high density, which are then merged into clusters (McInnes et al. 2017).

HDBSCAN offers several advantages over other clustering algorithms. It can handle clusters of different sizes and shapes, it can identify noise points, and it can automatically determine the optimal number of clusters without requiring a predetermined value. Furthermore, HDBSCAN does not require the specification of a fixed number of clusters, which can be challenging to determine in advance.

The HDBSCAN algorithm works by using an inexpensive estimate of density, known as the k th nearest neighbor. In simple terms, the goal is to find clusters in the data by identifying denser regions and differentiating them from sparser regions, which are considered noise. The density of points is estimated using the distance to the k th nearest neighbor, known as the core distance ($\text{core}_k(x)$) for a point x .

To make the clustering algorithm more robust to noise, a new distance metric called the mutual reachability distance is defined. The mutual reachability distance between two points a and b , denoted as $d_{\text{reach-}k}(a, b)$, is calculated as the maximum value among the core distances of a and b , as well as the original distance $d(a, b)$. In other words:

$$d_{\text{reach-}k}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\} \quad (1)$$

By using this metric, dense points (with low core distances) remain close to each other, while sparser points are pushed away to be at least their core distance away from any other point. This effectively separates denser regions from sparser regions.

This approach helps identify clusters by distinguishing denser regions from noise, making the clustering algorithm more robust to outliers and noisy data.

For our analysis, HDBSCAN was applied to the remaining 10951 sources using all five astrometry parameters, including galactic coordinates (ℓ , b), proper motion in R.A. and DEC., and parallax. However, we discarded the option to use Galactocentric coordinates (X , Y , Z) because converting between galactic coordinates and Galactocentric coordinates introduces uncertainties in the data, and HDBSCAN cannot handle errors.

2.2.1 HDBSCAN Hyperparameters [REVISAR]

The hyperparameters used for HDBSCAN were as follows: Euclidean distance metric (`metric="euclidean"`), and a minimal cluster size of 120 (`min_cluster_size=24`).

The minimal cluster size was chosen taking into account the number that produces at most 2 clusters. A smaller minimal cluster size value would split the hierarchical graph further, resulting in more nested solutions. On the other hand, a larger value would merge small groups and reduce the number of false-positive associations in the dataset. The remain hyperparameters weren't modified.

2.2.2 Parameter Derivation Across a Range of Membership Probabilities

To ensure a comprehensive characterization of NGC 6383, we extend our analysis to obtain cluster parameters over a range of membership probabilities. Specifically, we consider four probability thresholds: 60%, 70%, 80%, and 90%.

Using these probability thresholds allows us to explore the cluster's properties over a broader spectrum of member stars. By doing so, we aim to capture a more complete portrait of NGC 6383, encompassing not only the most probable members but also stars at the periphery of the cluster, which may have a lower likelihood of membership.

This approach aids in understanding how sensitive our derived cluster parameters are to the chosen membership probabilities. By comparing the parameters obtained at different probability thresholds, we can assess the robustness of our findings and better interpret their implications for the underlying structure and dynamics of NGC 6383.

This methodology ultimately provides a holistic view of the cluster, contributing to a more nuanced understanding of its composition and the various factors that influence it.

2.3 COSMIC and PyMC

In this study, we leverage the COSMIC (Characterization Of Star clusters using Machine-learning Inference and Clustering) program, developed by Lucas Pulgar-Escobar et al. (in prep.), to determine various characteristics of NGC 6383. COSMIC serves as a Swiss Army knife for astronomers, offering a comprehensive set of classes and functions built on Python 3.11. The program employs unsupervised machine learning algorithms to process large datasets, like Gaia data, and focuses on the application of clustering and Bayesian estimation to determine the fundamental parameters of open clusters such as structural parameters, reddening, age, and distance.

To bolster the Bayesian analysis in COSMIC, the PyMC package is utilized. PyMC⁴ is a Python package dedicated to Bayesian statistical modeling. It is particularly well-suited for applications that require advanced Markov Chain Monte Carlo (MCMC) and variational inference (VI) algorithms (Wiecki et al. 2023). Its flexibility

⁴ Available at <https://github.com/pymc-devs/pymc>

Figure 3. Center det.

and extensibility make PyMC an ideal choice for applying Bayesian analysis to complex problems, including the characterization of NGC 6383.

The PyMC library aims to modernize and simplify Bayesian modeling. It offers a user-friendly interface for setting the prior distribution, generally a Normal distribution, and executes the sampling process efficiently. One of the key advantages of using PyMC is its implementation of the No-U-Turn Sampler (NUTS), an advanced MCMC algorithm that allows for efficient exploration of high-dimensional parameter spaces.

The NUTS algorithm builds upon Hamiltonian Monte Carlo (HMC) techniques, simulating a physical system to generate proposals. These proposals are based on the topology of the posterior distribution, allowing for an adaptive and balanced exploration of the parameter space. The algorithm terminates when a U-turn is detected, which serves as an indicator that the space has been sufficiently explored.

This efficiency and adaptability give NUTS an edge over traditional methods like Metropolis-Hastings and Gibbs sampling, which are often constrained by the shape of the distribution or the dimensions being considered (Phan et al. 2019).

2.3.1 Parallax and Distance

Parallax measurements is crucial to determinate astronomical distances, providing a valuable alternative to relying solely on assumptions about the inherent properties of celestial objects. However, estimating distances from parallaxes is not a straightforward process due to the presence of measurement errors. In practice, converting a parallax into a distance involves an inference process that requires making assumptions and can introduce significant uncertainties, even when the errors are relatively small.

In the specific case of determining the distance to NGC 6383, we utilize parallax measurements and employ Bayesian inference techniques. A crucial aspect of this approach lies in establishing the prior distribution, which raises the fundamental question: *What is the most probable distance to the cluster?* Although this question may seem deceptively simple, careful consideration of the prior distribution is essential. Given that the cluster exhibits a denser concentration towards its center, it is reasonable to assume a priori that the distances of the cluster members follow a normal distribution.

Consequently, the probability distribution of a distance, denoted as d , given the observed parallax ϖ with uncertainty σ , can be expressed using the Bayes' theorem, as:

$$P(d | \varpi, \sigma) = \frac{1}{Z} P(\varpi | d, \sigma) P(d) \quad (2)$$

Here, Z represents the normalization constant:

$$Z = \int_0^\infty P(\varpi | d', \sigma) P(d') dd' \quad (3)$$

To model the measurements, We assume that for a star located at the true distance d_{true} , the true but unknown parallax is $\frac{1}{r}$. The measured parallax ϖ corresponds to a noisy estimation of $\frac{1}{r}$. We adopt a normal distribution to represent the distribution of ϖ , with an unknown mean of $\frac{1}{r}$ and a known standard deviation of σ . In other words, we assume that ϖ is drawn from the following distribution:

Figure 4. This figure presents the distribution of proper motion for the cluster, depicted by an ellipse. The ellipse represents the mean proper motion $\bar{\mu}$ and the covariance matrix (Σ), reflecting the center and shape of the proper motion distribution, respectively. The color of each point in the scatter plot corresponds to the assigned probability of the source. The gray error bars indicate the measurement uncertainties in the proper motion coordinates. The x -axis represents the corrected proper motion in the α^* direction (μ_{α^*}), while the y -axis represents the corrected proper motion in the δ direction (μ_δ). The green star marker and error bars represent the proper motion coordinates and their uncertainties for the source HD 159176. The ellipse encapsulates the estimated 95% region of the proper motion distribution.

$$P(\varpi | d, \sigma) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left[-\frac{\left(\varpi - \frac{1}{d}\right)^2}{2\sigma^2} \right] \quad (4)$$

A priori, d follows a truncated normal distribution with a lower limit of 0, since $d > 0$. We utilize frequentist estimates for the mean and standard deviation, determined through propagation formulas. Specifically, the mean distance $\bar{d}$ is estimated to be 1.1 kpc, with a corresponding standard deviation of $\sigma = 0.011$ kpc.

Figure A8 depicts the graphical representation of the parallax model implemented.

2.4 Proper Motions

To model the distribution of proper motions within the cluster, we assume a two-dimensional normal distribution. This choice is based on the understanding that the motions of individual cluster members in the plane of the sky exhibit both systematic and random components. By using a 2D normal distribution, we can effectively capture the typical motion patterns and dispersion of proper motions observed among the cluster members.

The two-dimensional normal distribution provides a mathematical framework to describe the probability density function (PDF) of proper motions in both the right ascension (μ_α) and declination (μ_δ) directions. The PDF is characterized by two essential parameters: the mean vector (μ) and the covariance matrix (Σ). The mean vector represents the average proper motion values in each direction, while the covariance matrix describes the dispersion and correlations between the proper motions in different directions.

Estimating the mean vector and covariance matrix from the observed proper motion data allows us to determine the center and shape of the proper motion distribution, respectively. This information provides valuable insights into the overall motion behavior of the cluster, including any collective motions or substructures that may be present.

The multivariate normal distribution can be mathematically represented as:

$$p(\mu | \bar{\mu}, \Sigma) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp \left[-\frac{1}{2}(\mu - \bar{\mu})^T \Sigma^{-1}(\mu - \bar{\mu}) \right] \quad (5)$$

In this equation, $\mu = [\mu_\alpha, \mu_\delta]$ represents the vector of two-dimensional proper motion in right ascension and declination, $\bar{\mu} = [\bar{\mu}_\alpha, \bar{\mu}_\delta]$ denotes the mean vector, and Σ is the 2×2 covariance matrix. A priori, μ follows a normal distribution $\mathcal{N}(\mathbf{0}, \Sigma)$, where $\Sigma \sim \text{Exp}(1)$.

By fitting this multivariate normal distribution to the proper motion data, we can estimate the mean vector $\bar{\mu}$ and the covariance matrix Σ , enabling us to characterize the distribution of proper motions in the cluster along both the right ascension and declination axes.

The graphical model representation of the proper motion distribution is shown in Figure A7.

2.5 Determination of Cluster Center

To accurately determine the center of the cluster, we employ Kernel Density Estimation (KDE), specifically using the `scikit-learn`'s `KernelDensity` class. Unlike the Gaussian KDE, this function uses a more flexible approach, allowing for different kernel types and bandwidths to be considered.

To find the optimal parameters for KDE, we use Grid Search Cross-Validation (CV) over a range of bandwidths and kernel types. The bandwidths are tested over a range from 1×10^{-7} the minimum error of the positions and the maximum search cone, and the kernel types considered are Gaussian, Tophat, Epanechnikov, Exponential, Linear, and Cosine.

Finally, we identify the point with the maximum density value, which serves as the estimated center of the cluster. The function returns this center.

By using Grid Search CV and allowing for different kernels and bandwidths, this function offers a robust and adaptable way to locate the center of a star cluster. It accounts for the possibility of differing star distributions and densities, thereby providing a more accurate center estimation.

2.6 Radial Density Profile

To determine the tidal cluster radius, we utilize the density profile introduced by King (1962). This profile provides insights into the spatial distribution of stars within the cluster, and we consider two types: the 3-parameter King profile and the 2-parameter King profile. Both profiles capture the gradual decrease in stellar density from the cluster center to its outskirts.

The 3-parameter King profile is defined as follows:

$$f(R) = k \left[\frac{1}{\sqrt{1 + \left(\frac{R}{R_c}\right)^2}} - \frac{1}{\sqrt{1 + \left(\frac{R_t}{R_c}\right)^2}} \right]^2 + b \quad (6)$$

On the other hand, the 2-parameter King profile is given by:

$$f(R) = \frac{k}{1 + \left(\frac{R}{R_c}\right)^2} + b \quad (7)$$

In both equations, R represents the radius, R_c denotes the core radius, R_t is the tidal radius, k is a normalization constant related to the central density, and b is the background density.

The 2-parameter King profile is simpler compared to the 3-parameter profile but still provides valuable information about the density distribution within the cluster.

Both density profiles are useful for analyzing open clusters, especially in cases where the number of stars is too low to calculate demonstrative densities. The 2-parameter King profile offers more flexibility and can effectively capture density variations within the

Figure 5. This figure illustrates the density profiles of the cluster based on the 2-parameter and 3-parameter King models, utilizing the obtained parameters. The black points represent the observed density profile (RDP), with error bars denoting the associated uncertainties. The fuchsia curve represents the 3-parameter King model, while the blue curve represents the 2-parameter King model. The x -axis represents the radial distance (r) in arcminutes, and the y -axis represents the stellar density (ρ) in units of stars per square arcminute. The shaded region around the 3-parameter King model curve indicates the uncertainty range. Red dashed lines indicate the estimated core radius (R_c), tidal radius (R_t), and cluster radius (R_{cl}). The green dashed line represents the mean background density (b).

cluster region. By fitting both models to the observed data, we can assess their performance and determine which profile better describes the spatial distribution of stars in the cluster.

To calculate the density of stars per unit area, we divide the cluster region into concentric annuli. We use an annulus width equivalent to the KDE bandwidth found in the center determination for each probability, for radii ranging from 0 to half the maximum distance to the center, and then radii ranging to half the maximum distance to the center until the maximum distance. This approach allows us to analyze the radial variations in stellar density throughout the cluster region.

In our Bayesian analysis, we assume prior distributions for the parameters of the density profiles. The priors that we consider are as follows:

$$\sigma \sim \text{HalfNormal}(5)$$

$$k \sim \text{Normal}(5, 5)$$

$$R_c \sim \text{Normal}(2, 5)$$

$$R_t \sim \text{Normal}(20, 5)$$

$$b \sim \text{Normal}(0, 3)$$

These priors were selected based on a visual selection and following the radius described in Astorga et al. (2018). The graphical model representations of both the 2-parameter King profile and the 3-parameter King profile are shown in Figures A6 and A5.

By fitting these models to the observed density data, we can estimate the posterior distributions of the model parameters and make inferences about the radial density profile of the cluster.

2.7 AsTeCA

AsTeCA, or Automated Stellar Cluster Analysis, is a suite of tools designed to automate the standard tests applied to stellar clusters to determine their basic parameters. The software uses positional and photometric data to obtain precise and objective values for a cluster's center coordinates, radius, luminosity function, and integrated color magnitude, as well as characterizing its probability of being a true physical cluster.

AsTeCA incorporates a Bayesian field star decontamination algorithm and an isochrone fitting process to provide accurate estimates for a cluster's metallicity, age, extinction, and distance values. The software was validated on a large set of synthetic clusters and a smaller set of observed Milky Way open clusters Perren et al. (2015).

The software is written in Python and is available as an open-source code, making it accessible for use and modification by the broader scientific community.

To determine the age, AsTeCA fits the best isochrone to the Color-Magnitude Diagram (CMD) of NGC 6383. The process involves several steps, first AsTeCA generates a set of synthetic clusters, each

characterized by a unique combination of four parameters: metallicity (z), age, distance modulus, and extinction. These synthetic clusters are densely interpolated to contain 1000 points throughout their entire length, including the evolved parts. The synthetic clusters are then subjected to a maximum magnitude cut, an initial mass function (IMF) sampling, and a binary fraction assignment. The synthetic clusters are finally perturbed by a magnitude completeness removal function and an exponential error function, which are representative of the observed star cluster. Then, for each observed star in the cluster, AsTeCA calculates the probability that it belongs to a given synthetic cluster. This is done by comparing the observed star's magnitude and color with those of the synthetic cluster's stars. The probabilities for all observed stars are then combined to give the overall probability for the entire observed star cluster to have arisen from the synthetic cluster. The best fit between the observed star cluster and the set of synthetic clusters is identified by finding the synthetic cluster that produces the maximum overall probability. This is equivalent to searching for the maximum value in the 4-dimensional likelihood surface defined by the four parameters. Through this process, AsTeCA provides accurate estimates for a cluster's metallicity, age, extinction, and distance values. The method has been validated on a large set of synthetic clusters and a smaller set of observed Milky Way open clusters.

3 ANALYSIS OF THE RESULTS

The clustering algorithm returned two clusters and noise, as can be seen in ???. One of the detected cluster corresponded to an agglomeration of stars with similar proper motions, but is not a real cluster, due to the lack of physical, parallax agglomeration, and the velocity dispersion is over 1 km/s [here reference], therefore called *statistical cluster*. The other cluster, correspond to NGC 6383, in blue the high probability members (≥ 80 per cent), in brown the probably members (< 80 per cent).

Based in the membership probabilities, NGC 6383 has 247, 180, 113 and 64 members, over 60, 70, 80, 90 per cent of probability respectively, and according with the same probability, only 197, 149, 94, 61 members brighter than $G = 18$ mag.

The distribution of the sources' probabilities can be seen in Figure 1. The condensed cluster tree of the sources is shown in Figure ??, and the selection according to HDBSCAN is shown in Figure ??.

For brevity, only we are going to inform the results using the members with at least 70 per cent of probability, but the full set of results using different probability threshold, is in Table 3.

3.1 Positional and Motion Analysis of HD 159176

The cluster is located at a central position in equatorial coordinates of 263.6704 ± 0.0771 degrees in Right Ascension, and -32.5798 ± 0.0771 degrees in Declination, as can be seen in Fig. ??. As usual, this represent the projected position in the plane of equatorial coordinates, and is not the 3-D center of the cluster, but the determination of the cluster center is improved using the proper motion center and the probability of the stars.

The mean proper motion correspond to 2.5501 ± 0.0109 mas yr $^{-1}$ and -1.7261 ± 0.0092 in DEC, in equatorial coordinates, with a mean dispersion of 0.1471 and 0.12379, which is in the permitted values according with Perren et al. (2023); ?.

The obtained mean parallax is 0.922 ± 0.079 mas and the mean sampled distance is 1.096 ± 0.035 kpc. The values are slightly different from those obtained from **HERE COMPARISON**

In terms of radial velocities, we inferred an estimated mean radial velocity of 3.413 ± 32.766 using 15 sources with probability above 70 per cent, the relative large standard deviation is mainly due to the lack's of information of radial velocities presented in Gaia (tanto percent aqui citar), and this is reflected in that only ~ 8.3 per cent of the 70 per cent members of NGC 6383 has available Gaia data.

The binary fraction in star clusters is a key factor in understanding the effect of binary stars on the properties and dynamical evolution of the clusters. One of the main effects of a significant fraction on unresolved binaries in the central part of a star cluster is expected to be the production of large deviations in the cluster's proper motion distribution with respect to the single-star component since the resulting velocities may be much larger for the binary motions (Bica & Bonatto 2005). Due that the candidates to binary stars could be not selected during the clustering processes due that aren't part of the single star proper motion distribution, we selected stars with parallax between 3σ of the obtained mean parallax. We have plotted and fitted the projected velocity of members stars, with a mean projected velocity of 3.0825 ± 0.009 mas yr $^{-1}$. We only found one peak in the star's projected distribution, so we could not obtained the two peaks described in Bica & Bonatto (2005) as can be seen in **HERE IMAGE OF THE ANEXO.**, and consequently we could not obtain the binary fraction in the cluster.

3.2 Structural parameters

The cluster members extend outward up to ~ 23.4638 arcmin. The cluster has a tidal radius of 33.1275 ± 13.8725 arcmin, representing the maximum extent of the cluster. The core radius, indicating the densest region of the cluster, is 1.0781 ± 0.1110 arcmin. The parameters k and b , representing the cluster's density profile following a King model, are 18.1575 ± 1.6992 stars per square arcmin and 0.0217 ± 0.0122 stars per square arcmin, respectively. These results, visualized in Fig. 5, suggest the cluster exhibits distinct structural characteristics, with the concentration parameter $C = \log(R_t/R_c) \sim 3.4252$ highlighting a significant degree of central concentration compared to typical ranges found in literature [Harris 1996].

To explore the cluster's dynamical state and potential extra-tidal features, we calculated the Hill Radius (R_H) following the methodology outlined by Carrera et al. (2019):

$$R_H = R_{gc} \times \left(\frac{m_c}{3M_{gc}} \right)^{1/3}$$

where $R_{GC} = 7.2069$ kpc denotes the galactocentric distance, $m_c = 253.6338M_\odot$ the cluster mass, and $M_{gc} = 1.438 \times 10^1 M_\odot$ the Galactic mass within the cluster's orbit, obtained from

$$M_G = 2 \times 10^8 M_\odot \left(\frac{R_G}{30 \text{ pc}} \right)^{1.2}.$$

This calculation yields a significant observational extent of 18.8887 arcmin or 6.038 pc.

Moreover, the gravitationally bound radius was estimated using the method described by Pinfield et al. (1998), which considers the cluster mass and Oort constants. Applying the formula

$$r_g = \left(\frac{GM_{total}}{2(A-B)^2} \right)^{1/3},$$

where G is the gravitational constant, and A and B are the Oort constants, with values from Bovy (2017). For our cluster, the gravitationally bound radius is calculated as 10.06 ± 1.67 pc, equating to an angular size of 28.3409 ± 5.25 arcmin, considering the cluster's distance.

Figure 6. This comprehensive graphic showcases three different CMD of NGC 6383. In the left, the CMD of the cluster, that captures the essence of the cluster's stellar population and the best-fitting isochrone gracefully aligns with the CMD. Meanwhile, on the center side, a captivating Hess diagram depicts the comparison between observed and synthetic data, highlighting any variations and emphasizing the accuracy of the isochrone model. Finally, on the right section, the synthetic cluster takes shape, carefully integrated with the isochrone.

Tidal forces significantly impact the cluster's initial configuration and subsequent evolution. According with [AQUI REFERENCIA], the tidal radius, it's a measure of the gravitational balance between the cluster and galactic tidal forces, and based on the Galactic potential and assuming a galactic mass of $1.438 \times 10^{11} M_{\odot}$, the tidal radius is determined from

$$R_t = \left(\frac{M}{2M_g} \right)^{1/3} R_g,$$

with a deduced tidal radius of 24.13298 arcmin or 7.699 pc.

Our analysis revealed no significant extra-tidal structures beyond the calculated tidal, Hill, and gravitational bound radii. If present, such features are too faint for our current detection methods. The radial cumulative distributions of cluster members across various brightness levels indicate mass segregation, with brighter stars more centrally localized. This segregation suggests that low-mass members may mitigate the effects of mass concentration in the cluster's inner regions.

We also explored the density contrast parameter $\delta_c = 1 + \frac{f_0}{f_b}$ 312.3109, finding it below the threshold suggested by Bonatto and Bica (2009), hinting at the cluster's dynamical state. Furthermore, we estimated the limiting radius r_{lim} , defining the boundary where the cluster's density falls to background levels. This metric, crucial for understanding cluster boundaries and membership, resulted in a value consistent with expectations based on cluster mass and galactic environmental factors, the limiting radius is given by

$$r_{lim} = r_c \sqrt{\left(\frac{f_0}{3\sigma_{bg}} \right) - 1},$$

where σ_{bg} is the uncertainty of f_{bg} , obtaining 23.99400 arcmin.

3.3 Age, Extinction, and color-magnitude diagram

Fig. [tanto] show different CMDs of the cluster. A grid of PARSEC isochrones (Bressan et al 2012) was used by Asteca, as previously explained. The best median solution is an isochrone of age [edad], distance of [distancia], and $Z =$, after applying an extinction correction $A_G =$ and reddening correction $E(BP - RP) =$. The age and distance slightly differentiate from. The binary fraction used is..[here explain other selections of Asteca].

3.4 Photometric Properties of HD 159176

The luminosity function is the distribution of stars of a cluster in different magnitude bins. To construct the LF for NGC 6383, first we transformed the apparent G magnitudes into absolute magnitudes by using the parallax distance. This figure exhibits that the LF continues to increase up to...

The mass function is the distribution of masses of cluster membership per unit volume. A mass-luminosity connection can transform LF into mass function. Since we could not acquire an observational transformation, we must rely on theoretical model. To convert LF into MF, we used the clusters parameters derived in this paper and

the theoretical models given by Marigo et al (2017). The mass function slope can be derived from the linear relation

[ingresar formula]

In the above relation, tanto tanto.

The Salpeter value for the mass function slope is $x =$. This form of Salpeter demonstrate that the number of stars in each mass range declines rapidly with increasing mass. Our derived MF slope value [here] agrees with the Salpeter's slope within uncertainty. Utilizing this value of the mass function slope within the mass range 1-2.7 $M_{\odot}$, the total and mean mass for the cluster members were acquired as $\approx$ and $\approx$ respectively.

We distributed cluster members into three mass ranges as shown in.. this diagram exhibits that the cluster members show a mass segregation effect as bright stars seem to be more centrally concentrated than the low mass members. The plausible reason for this effect may be dynamical evolution, an imprint of star formation or both (dib et al 2008, allison 2009, pavlik 2020).

we have estimate the value of R_h (half-light radius) based on the transformation equation as given in larsen 2006.

3.5 Dynamical and Evolutional state

In the lifetime of star clusters, encounters between their members stars gradually lead to increased energy equipartition throughout the clusters. In this process, massive stars are concentrated towards the cluster core and transfer their kinetic energy to fainter stars. The relaxation time T_R is defined as the time in which the stellar velocity distribution becomes Maxwellian and is expressed by the following formula:

Using the above formula, the value of the dynamical relaxation time T_R is determined as [tanto], significantly less than the cluster's age. Hence, we conclude that NGC 6383 is a dynamically relaxed cluster.

According with [reference], the lifetime of an open cluster starting with a canonical IMF can be approximated to be

[aqui eq del link].

In this case, NGC 6383 will be dissolved within $T_{diss,0} = 2G_{yr}$.

We also calculated the dynamical evolution parameters $\tau = \frac{age}{T_E}$ is found to be much greater than 1..., which concludes that these objects are dynamically relaxed.

Also, the compact N-body stellar groups have to be eventually demolished tidally, either by the Galactic field or by the nearby transif of any giant molecular clouds (Converse Sthaler 2011). We have estimated the dissociation time of both the clusters using the relationship given by Binney Tremaine 2008

$$t_{dis} = 250 \left(\frac{M}{300} \right)^{\frac{1}{2}} \times \left(\frac{R_h}{2} \right)^{-\frac{3}{2}}$$

all object are very giht tan their relaxation times.

We have used the method given by inney Tremaine 1987 to estimate the dynamical relaxation time

$$T_E = \frac{N}{8 \log N} \times T_{cross}$$

where $T_{cross} = D/\sigma_V$ is the crossing time, D is the diameters, and σ_V is the velocity dispersion with a typical value of 3 km s⁻¹ according with sharma et al 2020 and bisht et al 2017.

3.6 Orbit of NGC 6383

[buscar alguie que sepa xd]

3.7 Comparative Analysis with Cluster Members

Compare HD 159176's properties with other cluster members. This could involve looking at age, mass, luminosity, and other stellar characteristics to argue for or against its membership in NGC 6383.

3.8 Binary cluster with Antalová 2

3.9 Discussion on Anomalous Findings

Discuss any anomalies in the data or unexpected findings, and their potential implications for the understanding of NGC 6383 and HD 159176.

A summary of the results are in the Table 2.

4 CONCLUSION

This study has not only provided valuable insights into NGC 6383 but has also underscored the potential of Bayesian analysis in the study of open clusters more broadly. By considering the full probability distributions of astrometric parameters, Bayesian analysis offers a robust and comprehensive approach to assessing cluster membership and characterizing uncertainties. This methodology opens up new avenues for future research, enabling a more detailed and nuanced analysis of open clusters in the Milky Way.

In conclusion, the application of Bayesian analysis to the study of NGC 6383 has yielded compelling results. The analysis supports the robustness of the analysis and strengthens the previous authors claims regarding the cluster's properties. The presence of this central binary system as a potential non-member highlights the need for caution when analyzing open clusters and considering the presence of chance alignments.

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APPENDIX A: EXTRA FIGURES

This paper has been typeset from a $\text{\LaTeX}$ file prepared by the author.

Parameter	Description	Value	Unit
Cluster center in RA c_α	Central position of the cluster in Right Ascension	263.6745	degrees
Cluster center in DEC c_δ	Central position of the cluster in Declination	-32.5763	degrees
Proper Motion RA μ_α	Angular motion of the cluster in Right Ascension	2.534 ± 0.014	mas yr^{-1}
Proper Motion DEC μ_δ	Angular motion of the cluster in Declination	-1.706 ± 0.011	mas yr^{-1}
Tidal Radius R_t	Maximum extent of the cluster	21.952 ± 4.447	arcmin
Cluster Radius $R_c l$	Outer boundary of the cluster	10.267	arcmin
Core Radius R_c	Radius within the densest region	1.123 ± 0.077	arcmin
Background Density b	Background density of stars	0.119 ± 0.062	stars arcmin^{-2}
Center density k	Center density of the cluster	10.267 ± 0.445	stars arcmin^{-2}
Tidal Radius R_t	Maximum extent of the cluster	21.952 ± 4.447	parsec
Cluster Radius $R_c l$	Outer boundary of the cluster	10.267	parsec
Core Radius R_c	Radius within the densest region	1.123 ± 0.077	parsec
Background Density b	Background density of stars	0.119 ± 0.062	stars parsec^{-2}
Center density k	Center density of the cluster	10.267 ± 0.445	stars parsec^{-2}
Parallax ϖ	Angular displacement of the cluster	0.9154 ± 0.034	mas
Distance (ϖ)	Distance to the cluster through parallax	1.0939 ± 0.039	kpc
Distance (Distance modulus)	Distance to the cluster through distance modulus	10.0230 ± 0.127	kpc
Mass M	Mass of the synthetic cluster used	565 ± 158	$M_\odot$
Extinction A_V	Attenuation of light due to interstellar dust	1.342	-
E(B-V)			mag
R_V			
Age	Age of the cluster	6.475 ± 0.180	log (age)
Metallicity z	Abundance of heavy elements in the cluster	0.0032 ± 0.0020	-
Distance Modulus $(m - M)_0$	Difference in apparent and absolute magnitudes	10.05 ± 0.27	Magnitude
Numbers of members	The numbers of stars that belong to the cluster	268	Count of stars
R_{GC}			
Total luminosity			
MF Slope			
Total mass			
Average mass			
Relaxation time			
Dynamical evolution parameter (τ)			

Table 2.

Figure A1. This figure displays the trace plot of the proper motion parameter μ for the Bayesian analysis. It provides insights into the posterior distribution of the proper motion parameter and allows for the assessment of convergence and uncertainty.

Figure A2. This figure presents the trace plot and summary statistics for the two-parameter King profile obtained from the Bayesian analysis. It offers a visual representation of the posterior distribution of the parameters, along with important statistical summaries.

Figure A3. This figure showcases the trace plot and summary statistics of the King profile parameters obtained from the Bayesian analysis. The trace plot illustrates the posterior distribution of the parameters, while the summary statistics provide key information such as mean, standard deviation, and credible intervals.

Figure A4. This figure depicts the equatorial coordinates and proper motions of the members of the NGC 6383 open cluster. The left panel shows the scatter plot of member positions in equatorial coordinates (right ascension, α , and declination, δ). The color-coded points represent the probability of cluster membership. Error bars indicate the uncertainties in the coordinates. The right panel displays the scatter plot of proper motions ($\mu_{\alpha} \text{ lpha}^*$ and μ_δ) for the cluster members. Again, the color of the points corresponds to the probability of membership, and error bars represent the uncertainties in the proper motions. The colorbar on the right indicates the probability values.

Parameter	Value at Different Probabilities				Standard Deviation				Unit
	60%	70%	80%	90%	60%	70%	80%	90%	
Probability									
Center in RA c_α	—	—	—	—	—	—	—	—	degrees
Center in DEC c_δ	—	—	—	—	—	—	—	—	degrees
Tidal Radius R_t	—	—	—	—	—	—	—	—	arcmin
Cluster Radius R_{cl}	—	—	—	—	—	—	—	—	arcmin
Core Radius R_c	—	—	—	—	—	—	—	—	arcmin
Background Density b	—	—	—	—	—	—	—	—	stars/arcmin ²
Center Density k	—	—	—	—	—	—	—	—	stars/arcmin ²
Parallax ϖ	—	—	—	—	—	—	—	—	mas
Distance (ϖ)	—	—	—	—	—	—	—	—	kpc
Distance (Distance Modulus)	—	—	—	—	—	—	—	—	kpc
Mass M	—	—	—	—	—	—	—	—	$M_\odot$
Extinction A_V	—	—	—	—	—	—	—	—	—
Age	—	—	—	—	—	—	—	—	log(age)
Metallicity z	—	—	—	—	—	—	—	—	—
Distance Modulus $(m - M)_0$	—	—	—	—	—	—	—	—	magnitude
Number of Members	—	—	—	—	—	—	—	—	count
PM in RA μ_α	—	—	—	—	—	—	—	—	mas/yr
PM in DEC μ_δ	—	—	—	—	—	—	—	—	mas/yr

Table 3. Detailed parameters of the open cluster NGC 6383 at different probability levels. The table includes structural parameters such as tidal, cluster, and core radii, and star densities, among others. Each parameter is reported at 60%, 70%, 80%, and 90% probabilities, along with their respective standard deviations.

Figure A5. Graphical model representation of the 3-parameter King profile for the density distribution within the cluster.

Figure A6. Graphical model representation of the 2-parameter King profile for the density distribution within the cluster.

Figure A7. Graphical model representation of the proper motion distribution within the cluster.

Figure A8. Graphical representation of the parallax model implemented for NGC 6383, showing the prior distribution of distance d and the likelihood function $P(\varpi | d, \sigma)$.